Brewbeacon

BREWBEACON: EXECUTIVE SUMMARY

Problem

Home brewing is a hobby built on patience, and its biggest frustration is inconsistency. A brewer can follow the same recipe twice and get two very different beers. The usual causes are invisible: fermentation temperature drifting overnight, yeast that stalls halfway, or a brewer bottling before the beer has actually finished. Today most hobbyists check progress by pulling a sample, floating a hydrometer in it, and reading a number by eye. Each sample risks contamination and wastes beer, so most brewers check only once or twice and then guess.

We surveyed 212 home brewers through two regional brewing clubs and online hobbyist groups. Sixty-eight percent said they had thrown out or been disappointed with at least one batch in the past year because of a fermentation problem they did not catch in time. Fifty-four percent said they do not measure gravity during fermentation at all. Brewers told us they wanted reassurance and direction, not just data. That gap is the opportunity.

Solution

Brewbeacon is a compact sensor that clips onto the outside of a standard fermenter and measures gravity and temperature continuously, sending readings to a mobile app every fifteen minutes. Because it does not touch the beer, there is no contamination risk and no wasted sample.

The app is where the product becomes distinctive. A brewer selects a recipe, and Brewbeacon builds an expected fermentation curve. As real readings arrive, the app compares them to that curve and gives specific coaching: that the beer is running two degrees warm and should be moved somewhere cooler, that fermentation looks finished and it is safe to bottle, or that a stall suggests the yeast needs attention. Over time the app learns each brewer's setup and recipes, so advice improves with every batch. The result is what brewers actually want: the same good beer, again and again.

Market

There are an estimated 1.1 million active home brewers in the United States, based on industry hobbyist surveys, and a larger group of casual and aspiring brewers who buy starter kits each year. Brewers are an unusually committed customer. Our survey respondents reported spending an average of 380 dollars per year on the hobby, and many already buy accessories and upgrades.

Our initial target is the roughly 300,000 brewers who brew at least eight batches a year and already own a dedicated fermenter, since they feel the cost of failed batches most. If we reach even five percent of that group, that is 15,000 customers. At our planned pricing, this represents over 1.5 million dollars in hardware revenue and a recurring subscription base of more than 500,000 dollars a year. The same sensor concept extends naturally to related fermentation hobbies such as cider, mead, kombucha, and wine, which would widen the market considerably.

Business Model

Brewbeacon earns money in two ways. First, we sell the sensor for 69 dollars, with a unit cost of about 22 dollars at our planned first production volume, giving a gross margin of roughly 68 percent on hardware. Second, we offer an optional coaching subscription at 4 dollars a month or 36 dollars a year, which unlocks recipe-based curves, personalized advice, and batch history. The sensor works with basic charting for free, so the subscription is a genuine upgrade rather than a paywall.

Subscriptions matter because they turn a one-time purchase into recurring revenue with very low marginal cost. In our current group of users, 24 of the 31 paying customers have chosen the subscription, a conversion rate of about 77 percent among buyers. We also plan modest affiliate

revenue from recipe kits and ingredient suppliers, though we do not rely on it in our projections.

Competition

There are two kinds of alternatives. The first is the manual hydrometer, which is cheap but tedious, risks contamination, and gives only occasional snapshots. The second is a small number of existing electronic fermentation monitors. These tend to cost two to four times as much as Brewbeacon, often require the device to float inside the beer, and mostly display numbers without telling the brewer what to do. Several are aimed at serious or semi-professional brewers rather than the broad hobbyist base.

Our advantage is the combination of three things: a non-contact clip-on design, a low price point, and coaching rather than raw data. Coaching is also the hardest part for a competitor to copy, because it improves as we gather more batch data. Every batch our users brew makes the advice better for everyone, which builds a data advantage over time.

Team

Maya Okonkwo-Reyes, chief executive, is a senior studying business and a lifelong homebrewer who has brewed over sixty batches. She leads strategy, customer research, and partnerships. Daniel Thackeray, chief technology officer, is a computer engineering student who designed the sensor hardware and firmware. Priya Venkataraman, head of data, is a statistics graduate student who built the fermentation models behind the coaching engine. Luis Ferreira-Nakamura, head of product and design, is a design student who created the app and the clip mechanism. We also have informal guidance from a faculty advisor in food science at our university and two experienced brewing club members who act as test brewers.

Milestones and Financials

Over the past year we defined the concept, built a working prototype, and placed it with our first external users. Forty-two home brewers have used Brewbeacon for ten weeks across 118 batches. In that group, 31 have paid, bringing in 2,139 dollars in hardware sales and 1,290 dollars in annualized subscription revenue. Users reported that Brewbeacon helped them catch a problem in 37 percent of batches, and 88 percent said they would be disappointed if they could no longer use it.

Our prototypes are currently assembled by hand at a cost of about 58 dollars per unit. A first production run of 1,000 units would reduce this to about 22 dollars. We project 400 sensors sold in the first twelve months after launch, growing to 3,500 in year two, with subscription revenue rising from about 6,000 dollars in year one to over 60,000 dollars in year two. We expect to reach break-even on operating costs late in year two. We have not yet formed a legal entity and plan to incorporate once we secure funding.

What We Are Asking For

We are seeking 40,000 dollars. About 24,000 dollars would fund the first production run of 1,000 units, including tooling and certification testing. About 8,000 dollars would go to app development and cloud infrastructure, and the remainder to customer acquisition through brewing clubs and hobbyist communities and to legal costs of forming the company. Beyond funding, we would value mentorship in hardware manufacturing and introductions to brewing retailers and supply distributors. With this support, we believe Brewbeacon can become the standard tool that helps home brewers make the beer they intended, every time.